Exercises from Lectures 1–3. Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified.

Exercise 1. Graded algebras, duals, and non-graded modules

(1) For $A = \Bbbk A_n$ with $n \geq 2$, verify that $A_d = \operatorname{span}_\Bbbk\{e_{ij} \mid j - i = d\}$ defines a grading on A.

(2) Compute $\operatorname{rad}(\Bbbk A_n)$ from its definition as the intersection of all maximal right ideals.

(3) Let $A = \Bbbk A_3$, $M = e_{22}A$, and consider $\mathbb{D}(M)_d = \operatorname{Hom}_\Bbbk(M_{-d}, \Bbbk)$ with $(af)(m) = f(ma)$.

i. Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on it.

ii. Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that $\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$ as graded left A-modules.

iii. Deduce that $\mathbb{D}(M)$ is not isomorphic to any $(Ae_{jj})(r)$, $j \in \{1,2,3\}$, $r \in \mathbb{Z}$.

(4) **Modules that are not graded.** Let A be the algebra of matrices

$$X = \begin{pmatrix} a & u\alpha + v\beta \\ 0 & b \end{pmatrix} \qquad (a, b, u, v \in \Bbbk),$$

with the usual matrix product, where scalars act on the space $\Bbbk\alpha \oplus \Bbbk\beta$ in the obvious way:

$$\begin{pmatrix} a & p \\ 0 & b \end{pmatrix}\begin{pmatrix} a' & p' \\ 0 & b' \end{pmatrix} = \begin{pmatrix} aa' & ap' + pb' \\ 0 & bb' \end{pmatrix} \qquad (p, p' \in \Bbbk\alpha \oplus \Bbbk\beta).$$

Grade A by $\deg\alpha = 1$, $\deg\beta = 0$, with diagonal entries in degree 0. For $(\lambda, \mu) \in \Bbbk^2$ let $M(\lambda, \mu) = \Bbbk^2$ (row vectors) with right action

$$(x_1, x_2) \cdot X = (ax_1,\ (\lambda u + \mu v)x_1 + bx_2).$$

i. Verify that A is a graded algebra and that $M(\lambda, \mu)$ is a right A-module for every $(\lambda, \mu) \in \Bbbk^2$.

ii. Prove that $M(\lambda, \mu)$ admits a grading if and only if $\lambda\mu = 0$.

(5) **Independence of the symmetrising form.** Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ and $M = e_{11}A$ (inherited gradings). For $c = (c_1, c_2) \in (\Bbbk \setminus \{0\})^2$, consider the symmetrising form

$$\tau_c(ae_{11} + be_{22}) = c_1 a + c_2 b.$$

Recall that we have a the natural isomorphism $\alpha_c : (-)^{(*)} \xrightarrow{\sim} \mathbb{D}$ assocaited to τ_c given by $f \mapsto \tau_c \circ f$, which induces a left A-action $\cdot_c$ on $M^{(*)}$ induced by α_c is given by

$$a \cdot_c f = \alpha_c^{-1}(a\,\alpha_c(f)), \qquad (a\lambda)(m) = \lambda(ma).$$

Let

$$\varphi_0(ae_{11} + be_{12}) = ae_{11}, \qquad \varphi_1(ae_{11} + be_{12}) = be_{22}.$$

i. Check that φ_0, φ_1 forms a homogeneous basis of $M^{(*)}$ (specify their degrees). Compute $e_{11} \cdot_c \varphi_i$, $e_{22} \cdot_c \varphi_i$, $e_{12} \cdot_c \varphi_i$ for $i \in \{0,1\}$. Prove that $e \cdot_c \varphi_i = e \cdot_{c'} \varphi_i$ for all $e \in \{e_{11}, e_{22}, e_{12}\}$ and $i \in \{0,1\}$ if and only if $(c_1', c_2') = \rho(c_1, c_2)$ for some $\rho \in \Bbbk \setminus \{0\}$.

ii. More generally, let A be a locally finite-dimensional graded algebra with semisimple $A_0 = S$, and τ, τ' be symmetrising forms on S with assocaited natural isomorphisms α_τ and left A-action $\cdot_\tau$. Prove that there is an invertible $z \in S$ with $\tau'(s) = \tau(zs)$ for all s, and that

$$\gamma : (M^{(*)}, \cdot_{\tau'}) \to (M^{(*)}, \cdot_\tau), \quad (\gamma f)(m) = z\, f(m),$$

is an isomorphism of graded left A-modules. Deduce that in (1) the two modules $(M^{(*)}, \cdot_c)$ and $(M^{(*)}, \cdot_{c'})$ are isomorphic for all c, c'.
Hint. Compare $\alpha_{\tau'}$ with α_τ.

Exercise 2. Tensor products, S-duals and quadratic duals
Let S be a finite-dimensional semisimple $\Bbbk$-algebra. Let V be a finite-dimensional right S-module and W a finite-dimensional S-bimodule, and define

$$\theta_r : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*, \qquad \theta_r(g \otimes f)(v \otimes w) = g(f(v)w).$$

(1) Prove that θ_r is well defined: check bilinearity in (g, f) and in (v, w), invariance under $vs \otimes w \leftrightarrow v \otimes sw$ and $gs \otimes f \leftrightarrow g \otimes sf$, and right S-linearity of $\theta_r(g \otimes f)$. When V is a bimodule, show $\theta_r(sg \otimes ft) = s\,\theta_r(g \otimes f)\,t$.

(2) Prove that θ_r is bijective.

(3) Let $p : V \to V'$ be right S-linear and $q : W \to W'$ a bimodule map. Prove $(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*)$.

(4) Let $S = \Bbbk^{\oplus 3}$ with idempotents e_1, e_2, e_3, $V = \Bbbk a$ with $e_1 a e_2 = a$, and $W = \Bbbk c$ with $e_2 c e_3 = c$. Compute $W^* \otimes_S V^*$, $V^* \otimes_S W^*$ and $(V \otimes_S W)^*$, and describe θ_r. What happens if the two duals are tensored in the original order?

(5) Suppose instead that $S = \Bbbk[x]/(x^2)$ and $V = W = S/(x)$, with the same definitions of U^* and θ_r. Show that $W^* \otimes_S V^*$ and $(V \otimes_S W)^*$ are both 1-dimensional but $\theta_r = 0$. At which step of your proof of (2) does the argument fail?

(6) **A square with a parameter.** Let $S = \Bbbk e_1 \oplus \Bbbk e_2 \oplus \Bbbk e_3 \oplus \Bbbk e_4$ and let V be an S-bimodule with basis a, b, c, d and

$$e_1 a e_2 = a, \quad e_2 b e_4 = b, \quad e_1 c e_3 = c, \quad e_3 d e_4 = d,$$

and all other idempotent actions zero. The only composable words of length 2 are ab and cd. For $\lambda \in \Bbbk$ set $A_\lambda = T_S(V)/(ab - \lambda cd)$. Using θ_r, find the relation space $R^\perp$ and a presentation of $A_\lambda^!$ for every λ, including 0. Then dualise once more to recover the relation $ab - \lambda cd$.
Hint. Evaluate $\mu b^* a^* + \nu d^* c^*$ on $ab - \lambda cd$.

(7) **A monomial example.** Let $B = T_\Bbbk(\Bbbk x \oplus \Bbbk y)/(x^2)$ with $\deg x = \deg y = 1$, $S = \Bbbk$. Compute $B^!$ (with $X = x^*$, $Y = y^*$) and find a homogeneous basis of $B^!$.

Exercise 3. Quadratic presentations and Hilbert series
The parts are independent.

(1) Let $n \geq 3$, $Z \subset \{1, \dots, n-2\}$, and $A_Z = \Bbbk A_n/(e_{k,k+2} \mid k \in Z)$ with the quotient grading.
   i. Show that $\{e_{ii}\} \cup \{e_{ij} \mid i < j, \text{ no } k \in Z \text{ with } i \leq k \leq j-2\}$ is a basis of A_Z.
   ii. Write $A_Z = T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.

(2) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and let $V = \Bbbk a \oplus \Bbbk b$ be an S-bimodule with $e_1 a e_2 = a$, $e_2 b e_1 = b$ and all other idempotent actions zero. Find a basis of $T_S(V)_d$ for all $d \geq 0$ and compute $h_{T_S(V)}(t)$. Then find a basis of $T_S(V)/(ab)$ and explain why it is finite-dimensional.
   Hint. Count alternating words in each degree, including degree 0.

(3) Let V, W have bases $x_1, \dots, x_n$ and $y_1, \dots, y_m$, generators in degree 1, and consider $\mathsf{S}(V)$ and $\bigwedge(W)$. Show that
$$\dim_{\Bbbk} \mathsf{S}(V)_2 = \tfrac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\textstyle\bigwedge W)_2 = \tfrac{m(m-1)}{2}.$$

(4) **Counting words.** Let $B = T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)/(x^2)$ be the algebra of Exercise 2(7).
   i. Describe a basis of B_d, and compute $a_d = \dim B_d$ for $0 \leq d \leq 6$.
   ii. Express a_d through a_{d-1}, a_{d-2} by sorting words by first letter, and find $h_B(t)$ as a rational function. Using the basis of $B^!$ from Exercise 2(7), compute $h_{B^!}(t)$ and verify $h_B(t) h_{B^!}(-t) = 1$.

(5) Let A_λ ($\lambda \in \Bbbk$) be the algebra of Exercise 2(6). Prove that $A_\lambda \cong A_\mu$ as graded algebras over S if and only if $\lambda = \mu = 0$ or both are non-zero.
   Hint. A graded isomorphism over S sends each of a, b, c, d to a scalar multiple of itself.

Exercise 4. Computing quadratic duals
The parts are independent.

(1) Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1 with $C_{ij} = \dim_{\Bbbk} e_i V e_j$, and let $A = T_S(V)/(V \otimes_S V) = S \oplus V$.
   i. Prove $A^! = T_S(V^*)$, $h_A(t) = I + tC$, and $h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}$.
   ii. Deduce $h_A(t) h_{A^!}(-t)^{\mathsf{T}} = I$.

Hint. For (i) use $V \otimes_S V$ as the relation space and the formula $h_{T_S(V)}(t) = (I - tC)^{-1}$ of Lecture 2.

(2) For $q \in \Bbbk$ let $C_q = \Bbbk\langle x, y\rangle/(x^2, y^2, xy + qyx)$, $\deg x = \deg y = 1$.
   i. Find a homogeneous basis of C_q and compute all products of two basis elements, treating $q = 0$ and $q \neq 0$. Justify linear independence of your basis.
   ii. For $u = \alpha x + \beta y$ compute u^2. For each q, determine the (α, β) with $u^2 = 0$, and the q for which every degree-1 element squares to zero.
   iii. Prove: $C_q \cong C_{q'}$ as graded algebras if and only if either $q = q'$ or $qq' = 1$. Deduce that $C_1 \ncong C_{-1}$ when $\operatorname{char} \Bbbk \neq 2$ (they differ already as ungraded algebras, since one is commutative), and say what changes in characteristic 2.
   iv. Compute a quadratic presentation of $C_q^!$ using θ_r, and its Hilbert series. Verify $h_{C_q}(t) h_{C_q^!}(-t) = 1$. Compare $C_0^!$ with $C_1^!$.

Hint for (3). A graded isomorphism is determined by a linear bijection of degree-1 parts and preserves the set of lines $\Bbbk u$ with $u^2 = 0$. For (4), evaluate a general combination of X^2, XY, YX, Y^2 on x^2, y^2, and $xy + qyx$, with $X = x^*$, $Y = y^*$.